BundleLM: Bundle Construction Using Language Model

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Abstract. Bundle construction is inherently a multimedia composition task: constructing an outfit or playlist requires jointly analyzing heterogeneous visual, textual, and audio content to determine cross-item compatibility and bundle-level completeness. Existing methods typically reduce this task to item-level prediction or fixed-size completion, limiting their ability to model comprehensive coherence, compositional completeness, and adaptive termination. To address these limitations, we propose BundleLM, a language-model-inspired framework that formulates multimedia bundle construction as autoregressive generation in the bundle space. BundleLM first fuses multimodal content features, collaborative signals, and ID embeddings into unified item representations. It then employs a causal Transformer decoder to generate items step by step, where an explicit end token together with early- and late-stopping penalties enables length-adaptive and structurally complete generation. Finally, we introduce a multi-reward GRPO alignment stage to optimize bundle-level quality from semantic similarity, item overlap, and length alignment perspectives. Experiments on the POG fashion dataset and the Spotify playlist dataset show that BundleLM consistently outperforms representative baselines across multiple evaluation metrics, validating the effectiveness of the autoregressive bundle generation paradigm and the proposed multi-reward alignment mechanism.

Keywords: Multimedia Bundle Construction, Multimodal Representation Learning, Reinforcement Learning

1 Introduction

Product bundling is a long-standing and effective strategy in recommender systems, aiming to automatically construct semantically coherent and functionally complete item sets tailored to user preferences [20, 15]. More fundamentally, bundle construction requires multimedia content analysis and compositional reasoning: constructing a satisfactory bundle often involves jointly understanding heterogeneous content, such as product images and textual attributes in fashion or audio characteristics and metadata in music, while modeling compatibility among multiple items. It has been widely applied in domains such as outfit recommendation [4] and music playlist generation [3], benefiting both user experience and business operations [16, 14]. Traditionally, bundles were curated manually; however, the explosive growth of items has rendered this approach impractical, leading to a growing interest in automated methods. Moreover, current evaluation metrics also fail to reflect the holistic quality of bundles, further limiting progress in this direction.

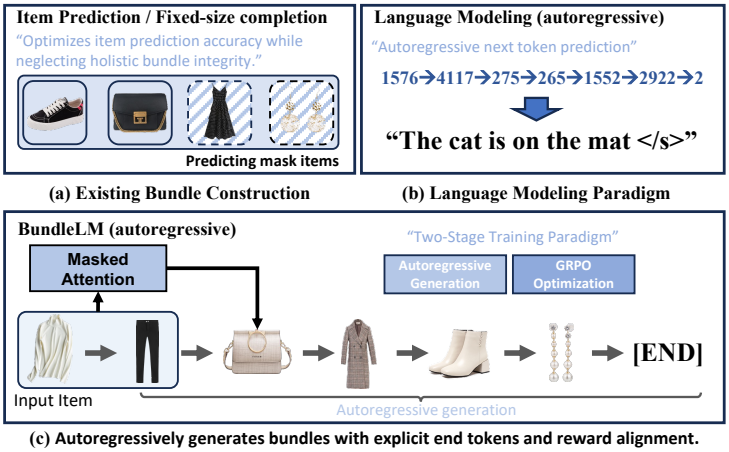


Fig. 1. Existing methods reduce bundle construction to item-level prediction (a). Inspired by language modeling (b), BundleLM generates bundles autoregressively with an explicit end signal, ensuring coherence and completeness (c).

Existing bundle construction approaches typically comprise two components: bundle representation learning, which captures the semantics and structure of a partial bundle, and item representation learning, which encodes individual items for compatibility matching. Some methods learn bundle–item compatibility in a shared representation space [16], while others use multimodal large language models for prompt-based prediction or multiple-choice selection [13]. Despite these differences, most methods formulate bundle construction as conditional item selection, either adding items to a partial bundle [16, 13] or generating a predefined number of items [6, 27]. Even when modeling dependencies among selected items, they do not explicitly learn whether the bundle is complete; instead, its size is externally specified or determined by the evaluation protocol. Consequently, their objectives emphasize item-level matching rather than the semantic coherence and structural completeness of the entire bundle. A satisfactory bundle, however, should terminate at an appropriate, input-dependent length, such as when an outfit becomes stylistically complete or a playlist semantically coherent. Without explicit termination, generation may stop prematurely or continue beyond completion, motivating a self-terminating formulation.

Autoregressive language models [1] generate complete sequences by conditioning each token on its preceding context. Adopting this paradigm for bundle construction, however, is not simply replacing an existing backbone with a sequential decoder. Conventional sequential approaches typically repeat next-item prediction for a predefined number of steps and are therefore not explicitly trained to recognize when the bundle becomes complete. In contrast, we treat the entire construction trajectory, including an explicit end token, as the learning target. Although a bundle is ultimately evaluated as a set, its generation can be factorized sequentially, with each item conditioning subsequent choices and the end token marking an input-dependent completion boundary. As shown in Figure 1, this formulation enables the model to learn both what to gener-

ate and when to terminate. It shifts the focus from isolated next-item accuracy to bundle-level semantic coherence, functional complementarity, and structural completeness while supporting variable-length generation. Thus, item selection, completion, and termination are learned jointly and evaluated holistically.

To address the aforementioned issues, we propose BundleLM, a bundle construction framework grounded in the language modeling paradigm, which alleviates the limitations of existing methods in semantic coherence and structural integrity. Our approach follows a progressive two-stage training paradigm. In the first stage, we perform supervised fine-tuning (SFT) on real-world bundles and introduce a causal mask to enable autoregressive bundle generation. An explicit end token and length constraints allow the model to learn an appropriate termination mechanism while avoiding degenerate cases of overly short or excessively long bundles. In the second stage, initialized from the SFT-trained model, we adopt Group Relative Policy Optimization (GRPO) [19] to further optimize bundle-level generation quality. The reward function consists of three complementary components: a semantic similarity reward computed by a pretrained bundle encoder, an item-overlap reward based on Jaccard similarity, and a length-alignment reward that encourages appropriate termination. Through the synergistic design of these components, BundleLM achieves coherent, reasonable, and self-contained bundle generation.

Extensive experiments on POG and Spotify demonstrate that BundleLM consistently improves bundle construction quality. Our main contributions are summarized as follows:

- To the best of our knowledge, we are the first to formulate bundle construction as a language modeling problem, factorizing complete bundle generation, including termination, into conditional item predictions.
- We propose BundleLM, which combines causal generation, explicit end-token supervision, length-aware stop control, and multi-reward GRPO alignment to optimize bundle coherence, item relevance, and length.
- Extensive experiments on POG and Spotify demonstrate SOTA performance, while ablation studies validate the effectiveness of the proposed stop-control and reward-alignment designs.

2 Related Work

Bundle Recommendation and Construction. While early studies focused on retrieving existing bundles [14], expanding item repositories have shifted the focus toward automatic bundle construction. Existing methods employ diverse mechanisms: latent factor models [12] for representation learning, sequential models [7] for item dependencies, graph structures [28] for complex relations, and attention architectures [24] for interactions. Recently, CLHE [16] introduced hierarchical multimodal encoding, while BundleMLLM [13] leveraged multimodal LLMs for prompt-driven item selection. Other works target personalized bundle generation [6, 21]. Despite these advances, most methods resemble open-ended sequential recommendation. Lacking explicit stopping criteria, they struggle to

capture the structural integrity and semantic coherence of complete bundles.

Language Models for Recommendation. Recommendation systems frequently adapt language models, utilizing RNNs [5], LSTMs [7], and Transformers [1] for sequential dependencies. Recently, pretrained LLMs have been widely employed for data generation [23], item ranking [2], multimodal representation [29], and generative recommendation [10]. Specifically for bundles, LLMs assist in multiple-choice selection [13] and intent alignment [21]. However, these approaches remain largely grounded in natural language, relying on linguistic transfer rather than capturing bundle-specific statistical structures (e.g., multimodal compatibility, collaborative relations, and set-level completeness). Instead of adapting a textual LLM, BundleLM instantiates autoregressive generation directly within the bundle space. By treating items as tokens, it factorizes complete bundle generation—including explicit termination—into conditional probabilities. Supported by multimodal/collaborative representations, autoregressive fine-tuning, and RL post-training, BundleLM operates as a bundle-native language model.

Reinforcement Learning for Bundle Task. Reinforcement learning effectively optimizes sequence-level objectives beyond the reach of maximum likelihood training [18], making it well-suited for interactive and sequential recommendation [30]. In bundle-related tasks, BYOB [6] uses RL for combinatorial optimization to select a predefined number of items, while BundleMCR [9] optimizes multi-turn conversational policies rather than directly generating complete sets. Although demonstrating RL's potential, these methods restrict their focus to fixed-cardinality selection or predefined dialogue actions, failing to explicitly optimize the step-by-step generation and adaptive termination of a complete bundle. To address this, BundleLM introduces a GRPO-based post-training stage on the autoregressive policy, directly optimizing complete bundles using holistic semantic-similarity, item-overlap, and length-alignment rewards.

3 Research Method

3.1 Problem Formulation

A bundle $b = \{i_1, \ldots, i_{n_b}\} \subseteq I$ is inherently an unordered set. To enable autoregressive generation, we sample a random permutation $\pi_b \sim \mathrm{Unif}(\Pi(b))$ and append a termination token e_{end}, yielding an ordered training target $\mathbf{y}_b^{\pi_b} = \langle i_{\pi_b(1)}, \ldots, i_{\pi_b(n_b)}, e_{\mathrm{end}} \rangle$. The likelihood is factorized sequentially as $p_\theta(\mathbf{y}_b^{\pi_b}) = \prod_{t=1}^{n_b+1} p_\theta(y_{b,t}^{\pi_b} \mid \mathbf{y}_{b,<t}^{\pi_b})$, where $\mathbf{y}_{b,<t}^{\pi_b}$ is the prefix at step t. We thus optimize the expected likelihood over all sampled bundle permutations:

$$\mathcal{L}_{\mathrm{AR}} = -\mathbb{E}_{b\sim\mathcal{D},\, \pi_b\sim\mathrm{Unif}(\Pi(b))} \left[\sum_{t=1}^{n_b+1} \log p_\theta \left(y_{b,t}^{\pi_b} \mid \mathbf{y}_{b,<t}^{\pi_b} \right) \right]. \tag{1}$$

At inference, an observed partial bundle x_b serves as the initial prefix. Causal masking ensures strict sequential dependency, and the model autonomously generates items until predicting e_{end} without requiring predefined length constraints.

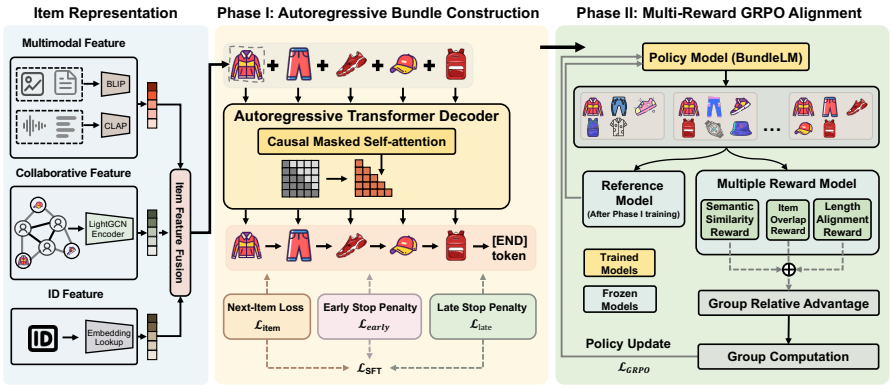


Fig. 2. Overview of BundleLM, which performs autoregressive bundle construction via stop-control losses based on fused multimodal and collaborative features, followed by multi-reward GRPO alignment.

3.2 Methods

Pretraining for Item and Bundle Representations. To establish unified semantics before sequential decoding, we pretrain representations for items and bundles. For each item $i \in I$, we extract textual features t_i and multimodal features m_i (image/audio) via BLIP [11] and CLAP [26], and collaborative filtering embeddings p_i via LightGCN [8]. We project their fusion along with an ID embedding $\mathbf{v}_i$ into a shared space $\mathbf{F}_i = [\text{avg}(t_i, m_i)\mathbf{W}_c; p_i\mathbf{W}_f; \mathbf{v}_i]$ and apply a multi-layer self-attention encoder to obtain the final item embedding $\mathbf{f}_i$. Bundle representations $\mathbf{e}_b$ are subsequently formed by aggregating constituent item embeddings through a bundle-level self-attention encoder.

During pretraining, we adopt a masked item prediction strategy: half of the items in each bundle are masked, and the model is trained to recover them conditioned on the remaining items. This denoising objective encourages the model to capture intra-bundle item dependencies and cross-modal compatibility.

Autoregressive Generation with Causal Mask. To enable step-by-step bundle generation, we apply an upper-triangular causal attention mask $\mathbf{M}_{\text{causal}}$ to the bundle self-attention layers, setting future position logits to $-\infty$. At step t, the context vector $\mathbf{e}_b^t = \text{avg}(\widetilde{\boldsymbol{E}}_b^{(Z)})$ aggregates the preceding $t-1$ items to predict the next item token, preventing information leakage from future elements.

Length-Aware Termination Learning. Autoregressive generation often suffers from premature stopping or excessive generation due to variable bundle lengths [17]. We resolve this through three synergistic objectives:

End Token Reweighting. We downscale the penalty for predicting the termination token in the cross-entropy loss:

$$\mathcal{L}_{\text{item}} = -\sum_{t=1}^{T} w_{y_t} \log p_\theta(y_t \mid \mathbf{e}_b^{t-1}), \quad \text{where } w_{y_t} = \begin{cases} \lambda_{\text{end}} \in (0,1), & y_t = \text{end} \\ 1, & \text{otherwise} \end{cases}. \tag{2}$$

Early-Stopping Margin Penalty. To prevent the termination logit $\ell_{t,\text{end}}$ from dominating before sequence completion, we enforce a margin $\delta > 0$ relative to the highest non-end logit $\ell_{t,\max}^{\neg\text{end}} = \max_{i \in I} \ell_{t,i}$:

$$\mathcal{L}_{\text{early}} = \mathbb{E}_{b \sim \mathcal{D}} \left[\frac{1}{|b|} \sum_{t=1}^{|b|} \max\left(0, \ell_{t,\text{end}} - \ell_{t,\max}^{\neg\text{end}} + \delta\right) \right]. \tag{3}$$

Late-Stopping Penalty. Conversely, to ensure termination upon generating the full ground-truth bundle $\mathbf{y}_{b,1:T_b}$, we penalize missing the end token at the sequence boundary:

$$\mathcal{L}_{\text{late}} = \mathbb{E}_{b \sim \mathcal{D}} \left[1 - p_\theta \left(e_{\text{end}} \mid x_b, \mathbf{y}_{b,1:T_b} \right) \right]. \tag{4}$$

The overall SFT objective combines these losses with balancing weights α and β: $\mathcal{L}_{\text{SFT}} = \mathcal{L}_{\text{item}} + \alpha \mathcal{L}_{\text{early}} + \beta \mathcal{L}_{\text{late}}$.

3.3 Multi-Reward GRPO for Bundle-Level Optimization

While SFT stabilizes termination, token-level likelihood optimization fails to directly enforce holistic criteria like semantic coherence and length alignment. To bridge this gap, we adapt Group Relative Policy Optimization (GRPO) [19] to bundle generation.

Given a partial bundle prefix x_b, the policy samples a group of G candidate bundles $\{\mathbf{y}_i\}_{i=1}^{G}$. Each candidate receives a scalar reward R_i, normalized into a group-relative advantage $\widehat{A}_i = (R_i - \mu_R)/(\sigma_R + \epsilon_A)$, where μ_R and σ_R denote the mean and standard deviation of rewards within the group. The GRPO objective optimizes the token-level policy ratio $r_{i,t}(\theta) = \frac{\pi_\theta(y_{i,t} | x_b, \mathbf{y}_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t} | x_b, \mathbf{y}_{i,<t})}$ against a frozen SFT reference policy π_{ref}:

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\mathbb{E}_{\substack{x_b \sim \mathcal{D} \\ \{\mathbf{y}_i\} \sim \pi_{\theta_{\text{old}}}}} \left[\frac{1}{G} \sum_{i=1}^{G} \frac{1}{T_i} \sum_{t=1}^{T_i} \Big(\min\left(r_{i,t}(\theta) \widehat{A}_i, \text{clip}(r_{i,t}(\theta), 1-\epsilon, 1+\epsilon) \widehat{A}_i \right) - \beta D_{\text{KL}}^{i,t} \Big) \right], \tag{5}$$

where T_i is sequence length, ϵ clips policy drift, and β controls KL divergence from π_{ref}.

Multi-Reward Design. To guide policy updates, we formulate a composite reward $R_i = w_s R_{\text{sem}} + w_o R_{\text{jaccard}} + w_l R_{\text{len}}$ comprising: (1) *Semantic Similarity (R_{sem})*: cosine similarity between the generated and ground-truth bundle embeddings via our pretrained bundle encoder; (2) *Item Overlap (R_{jaccard})*: Jaccard similarity against target items to penalize irrelevant selections; and (3) *Length Alignment (R_{len})*: penalty for cardinality deviation to discourage premature termination and over-generation.

Table 1. Statistics of the two real-world bundling datasets. "#" denotes the number of entities or interactions, and "Avg." denotes the corresponding average.

Dataset	#U	#I	#B	#B-I	#U-I	Avg. I/B	Avg. I/U
Spotify	118,994	10,284	12,270	216,393	6,642,657	17.63	55.82
POG	2,311,431	31,217	29,686	105,775	6,345,137	3.56	2.75

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on two multimodal benchmarks: POG fashion outfits [4] and Spotify playlists [3] (filtered to tracks with frequency ≥ 10 and playlist length ≤ 40). Both datasets are split into train, validation, and test sets at a $7:1:2$ ratio. Models are trained on full bundles and evaluated by generating continuations from a single seed item (statistics in Table 1).

Baselines. Following prior studies [16, 13], we compare BundleLM against representative bundle construction methods using identical multimodal features and training losses for fairness: **MultiDAE** [25]: an auto-encoder model using average pooling for item aggregation; **MultiVAE** [12]: a variational auto-encoder incorporating variational inference; **Bi-LSTM** [7]: a sequential model utilizing bidirectional LSTMs; **Hypergraph** [28]: a hyper-graph neural network modeling bundle interactions; **Transformer** [22]: an attention-based architecture modeling item-level dependencies; and **CLHE** [16]: a contrastive learning-enhanced hierarchical multimodal encoder. We evaluate all baselines under both top-k probability ranking and autoregressive decoding up to the maximum length.

Evaluation Metrics. Given a generated bundle $\hat{B}$ and its paired ground-truth B, we evaluate performance via three metrics: **Jaccard Similarity**: $\mathrm{Jac}_{\mathrm{single}}(\hat{B}, B) = \frac{|\hat{B} \cap B|}{|\hat{B} \cup B|}$, and its multi-reference extension $\mathrm{Jac}_{\mathrm{multi}} = \max_{B' \in \mathcal{R}(x)} \mathrm{Jac}(\hat{B}, B')$ matching against all valid references $\mathcal{R}(x)$ to handle multi-completion ambiguity. **Precision**: $\mathrm{Pre}(\hat{B}, B) = \frac{|\hat{B} \cap B|}{|\hat{B}|}$, measuring target item hit rate. **Bundle Semantic Similarity (BSS)**: $\mathrm{BSS}(\hat{B}, B) = \frac{\mathbf{h}_{\hat{B}}^{\top} \mathbf{h}_B}{\|\mathbf{h}_{\hat{B}}\|_2 \|\mathbf{h}_B\|_2}$, evaluating embedding-level consistency via our pretrained bundle encoder.

Hyperparameter Settings. Embedding dimension is fixed to 64, with single-layer GCN/Transformer modules. Models are trained with AdamW (lr=10^{-4}, weight decay=10^{-5}) up to 100 epochs. Batch sizes are 256 for POG and 64 for Spotify. Maximum generation lengths for autoregressive evaluation are set to 7 (POG) and 40 (Spotify). For SFT stop-control, we set end-token weight $\lambda_{\mathrm{end}} = 10^{-5}$, early-stop weight $\alpha = 5$, and late-stop weight $\beta = 10$. In GRPO post-training, group size $G = 4$, clipping $\epsilon = 0.2$, KL weight $\beta = 0.1$, with reward weights $w_s = 1.0$, $w_l = 0.1$, and $w_o = 10.0$.

Table 2. Overall performance comparison under autoregressive generation. Jac_m denotes the multi-reference Jaccard score Jac_{multi}, defined in Sec. 4.1.

	POG			Spotify		
Method	Jac_m	Prec.	BSS	Jac_m	Prec.	BSS
MultiDAE	0.061	0.034	0.731	0.013	0.015	0.428
MultiVAE	0.091	0.046	0.708	0.012	0.014	0.403
Bi-LSTM	0.102	0.049	0.735	0.012	0.014	0.376
Hypergraph	0.041	0.005	0.620	0.011	0.013	0.377
Transformer	0.111	0.049	0.727	0.026	0.029	0.489
CLHE	0.180	0.109	0.753	0.057	0.063	0.640
BundleLM	**0.250**	**0.158**	**0.755**	**0.059**	**0.065**	**0.652**
%Improv.	38.89%	44.95%	0.27%	3.51%	3.17%	1.87%

4.2 Overall Performance

Table 2 reports the overall comparison between BundleLM and baseline methods under a unified autoregressive setting. For a fair comparison, we adapt all baselines to the same autoregressive bundle construction protocol as BundleLM, where the model sequentially generates items starting from a given seed item. BundleLM achieves the best performance on both POG and Spotify, demonstrating the effectiveness of its multimodal item representation, stop-control objectives, and multi-reward alignment for bundle generation. The improvement is particularly significant on POG, where BundleLM substantially outperforms the strongest baseline in Jaccard_m and Precision. This may be attributed to the characteristics of POG bundles: fashion outfits are typically shorter and exhibit stronger compatibility, complementarity, and style consistency among items, making them well suited for autoregressive modeling. In contrast, Spotify playlists are usually longer and have weaker sequential dependency, making autoregressive generation more vulnerable to error accumulation and length-control issues. In Sec. 4.4, we further compare against top-k completion baselines to better understand the behavior of different generation paradigms across datasets.

4.3 Ablation Study

We conduct ablation studies to examine the stop-control objectives in SFT and the reward components in RL. As shown in Table 3, we use Jac_m as the primary metric to account for multiple plausible bundles associated with the same input. Removing either stopping objective degrades performance, with the early-stopping objective being particularly critical. Notably, on POG, removing only $\mathcal{L}_{\text{early}}$ causes a severe performance drop, whereas removing both stopping objectives partially restores it. This non-monotonic result suggests that the two objectives form a coupled control mechanism: $\mathcal{L}_{\text{late}}$ promotes termination at the target boundary, while $\mathcal{L}_{\text{early}}$ prevents this signal from inducing premature termination at shorter prefixes.

To further examine the termination behavior, we analyze the generated lengths on the POG test set. Among 5,938 predictions, approximately 65% emit end, in-

Table 3. Ablation study of BundleLM on POG and Spotify.

Stage	Variant	POG		Spotify	
		Jac_m	Prec.	Jac_m	Prec.
SFT	Full SFT	0.2403	0.1503	0.0583	0.0648
	w/o End Weight	0.1396	0.1309	0.0579	0.0640
	w/o Early&Late Stop	0.1940	0.1240	0.0577	0.0641
	w/o Early Stop	0.0524	0.0125	0.0578	0.0641
	w/o Late Stop	0.2341	0.1320	0.0577	0.0645
RL	w/o RL	0.2403	0.1503	0.0583	0.0648
	w/o BSS Reward	0.2426	0.1550	0.0584	0.0649
	w/o Length Reward	0.2431	0.1573	0.0579	0.0650
	w/o Jaccard Reward	0.2438	0.1568	0.0580	0.0653
	Full BundleLM	0.2502	0.1580	0.0590	0.0654

Table 4. Overall performance comparison of BundleLM and top-k baselines on POG and Spotify.

	POG			Spotify		
Method	Jaccard_m	Precision	BSS	Jaccard_m	Precision	BSS
MultiDAE	0.063	0.019	0.631	0.015	0.017	0.507
MultiVAE	0.091	0.046	0.708	0.014	0.016	0.477
Bi-LSTM	0.102	0.057	0.665	0.014	0.016	0.371
Hypergraph	0.056	0.015	0.689	0.025	0.030	0.623
Transformer	0.107	0.050	0.722	0.032	0.037	0.631
CLHE	<u>0.175</u>	<u>0.105</u>	<u>0.751</u>	<u>0.058</u>	**0.065**	<u>0.650</u>
BundleLM	**0.250**	**0.158**	**0.755**	**0.059**	**0.065**	**0.652**
%Improv.	42.85%	50.47%	0.53%	1.72%	–	0.31%

dicating that the model has learned an explicit termination mechanism. However, only around 19% match a valid reference length, while the premature-stop and maximum-length hit rates are approximately 19% and 35%, respectively. These results suggest that the main challenge lies not only in predicting end, but also in placing it at an appropriate position. The coexistence of premature termination and excessive generation further highlights the complementary roles of the two objectives: $\mathcal{L}_{\text{early}}$ suppresses premature termination, whereas $\mathcal{L}_{\text{late}}$ prevents unbounded generation.

The complete RL objective consistently outperforms the SFT-only model and all reduced variants. Removing any reward component leads to inferior performance, demonstrating their complementary roles: BSS captures bundle-level semantic quality, the length reward regulates output length, and the Jaccard reward encourages item-level agreement. Together, they provide effective bundle-level supervision for generation.

4.4 Impact of Generation Strategies

Table 4 presents results for the top-k completion strategy, where models directly select the highest-scoring items up to the target length. Comparing these

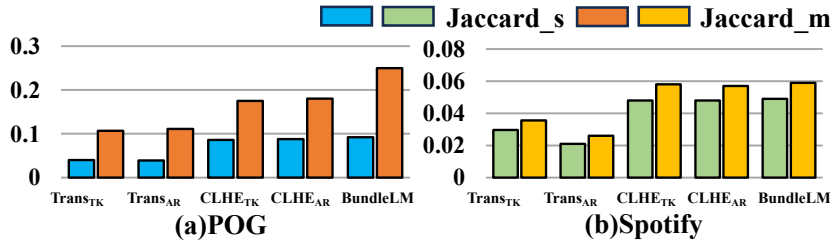


Fig. 3. Comparison between single-reference and multi-reference Jaccard evaluation under different generation strategies. TK denotes top-k completion, and AR denotes autoregressive generation.

with the results in Table 2, autoregressive generation performs better on POG, whereas top-k completion is more favorable on Spotify. This divergence stems from data characteristics: POG fashion outfits require strong stylistic and functional complementarity across steps, where autoregressive conditioning naturally aids sequential assembly. Conversely, Spotify playlists behave as unordered sets centered on co-occurrence, where top-k retrieval bypasses autoregressive error accumulation and length-control ambiguity. Across both paradigms, BundleLM consistently achieves leading or highly competitive performance.

4.5 Single- vs. Multi-Reference Evaluation

Unlike next-item recommendation, bundle completion is inherently a set-valued and one-to-many task: an input item can participate in multiple valid, compatible bundles. Evaluating against only a single paired ground-truth bundle may unfairly penalize plausible alternatives. To address this ambiguity, Fig. 3 compares single-reference (J_{single}) and multi-reference (J_{multi}) Jaccard scores, where J_{multi} measures the best match against the set of all valid candidate bundles $\mathcal{R}(x)$. Across all baselines and decoding strategies, J_{multi} consistently exceeds J_{single}, confirming that single-reference metrics underestimate bundle quality in one-to-many scenarios. Autoregressive decoding achieves comparable or superior multi-reference performance over top-ranked selection, verifying the utility of capturing sequential item dependencies. Furthermore, BundleLM achieves the highest scores under both protocols with pronounced gains under J_{multi}, demonstrating its capacity to generalize over broader compatibility distributions rather than merely overfitting to paired targets. We thus adopt J_{multi} as the primary set-level metric while retaining J_{single} for comparison with paired benchmarks.

4.6 Robustness to Input Prefix Size

To evaluate robustness to varying partial-bundle contexts, we conduct experiments on POG using two-item inputs. As shown in Table 5, BundleLM consistently outperforms Transformer and CLHE across all metrics, demonstrating that its advantages extend beyond the single-item setting. The two input settings introduce complementary challenges. A single-item prefix offers limited contextual information with higher selection uncertainty, whereas a two-item prefix provides richer context but leaves a shorter continuation window. Because evaluation

Table 5. Performance on POG with two-item input prefixes.

Method	Jac_s ↑	Jac_m ↑	Prec. ↑	BSS ↑
Transformer	0.052	0.056	0.059	0.870
CLHE	0.068	0.072	0.076	0.886
BundleLM	**0.078**	**0.084**	**0.113**	**0.888**

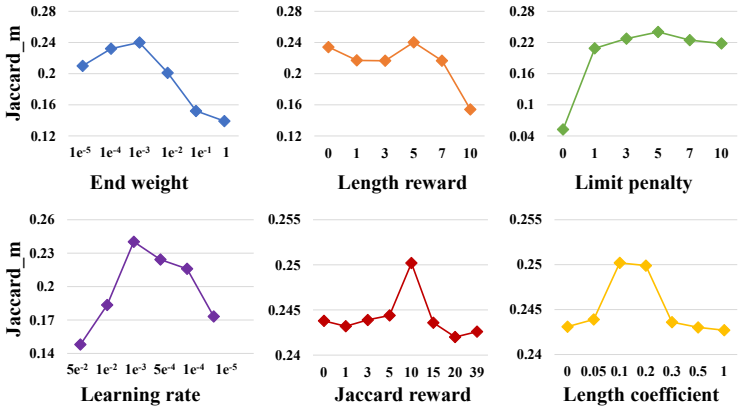


Fig. 4. The figure shows the hyperparameter sensitivity analysis of BundleLM on the POG dataset. The x-axis denotes different hyperparameter values, and the y-axis reports the Jaccard_m score on the validation set.

isolates the generated continuation, any misprediction or omission disproportionately impacts Jaccard overlap, while over-generation directly degrades Precision. Furthermore, a more constrained prefix narrows the valid multi-reference set and demands tighter termination control. Consequently, lower absolute overlap scores in the two-item setting reflect stricter metric sensitivity rather than degraded model capability. Across both configurations, the consistent gains confirm the robustness of BundleLM to different input prefix sizes.

4.7 Hyper-parameter Sensitivity

We conduct hyperparameter sensitivity experiments on POG, with the results shown in Figure 4. The SFT stop-control parameters substantially affect Jaccard_{multi}: moderate end-token and length-related weights balance bundle completeness and timely termination, whereas overly large values cause premature stopping or suppress useful item generation. Performance is also sensitive to the Jaccard-reward weight and length-alignment coefficient during GRPO, highlighting the need to balance item overlap, semantic consistency, and length. These results confirm the effectiveness of BundleLM's stop-control and multi-reward designs.

4.8 Qualitative Analysis (Case Study)

The case studies highlight that BundleLM generates more compact and semantically coherent bundles. In case (a), BundleLM differs from the ground truth by

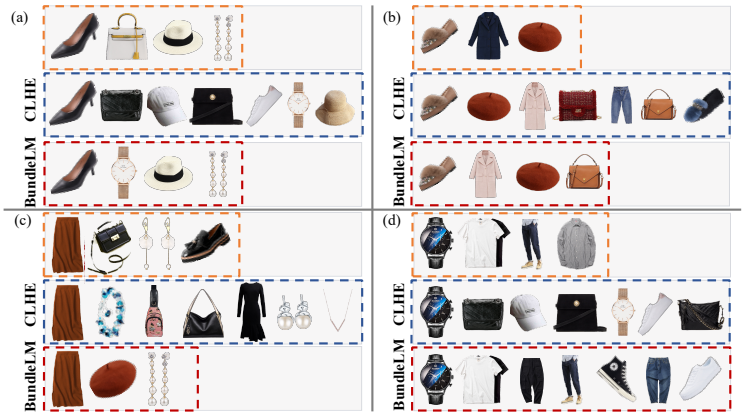


Fig. 5. The figure presents case studies on the POG dataset. In each case, the first row with an orange box denotes the ground-truth bundle, the second row with a blue box shows the result generated by CLHE, and the third row with a red box shows the result generated by BundleLM. All methods generate bundle completions from the same input item.

only one item and stops at an appropriate position, demonstrating precise item selection and length control. In case (b), while replacing the ground-truth coat with an alternative outer garment and adding a handbag, the resulting warm palette and coordinated accessories form a plausible alternative outfit. In case (c), although differing from the exact target items, the predicted items align well with the input seed in both color and style. These completions confirm that BundleLM generates compatible alternatives beyond rigid ground-truth matching, consistent with its bundle-level semantic and overlap rewards. Case (d) illustrates a remaining limitation: while maintaining a coherent menswear theme, the model fails to terminate promptly and outputs redundant item categories, suggesting potential for stronger redundancy suppression. In contrast, CLHE lacks explicit termination mechanisms, frequently yielding repetitive or incompatible items across all scenarios.

5 Conclusion and Future Work

This paper presented BundleLM, an autoregressive framework for bundle construction. Departing from fixed-cardinality ranking, BundleLM generates items sequentially with autonomous termination via an explicit end token and length-aware stop-control objectives. Multi-reward GRPO further enforces bundle-level quality by jointly optimizing semantic similarity, item overlap, and length alignment. Experiments across POG and Spotify confirm that BundleLM consistently outperforms state-of-the-art baselines, with ablation and case analyses validating the necessity of its dual-stage training design.

In future work, we will explore cross-domain bundle generation to uncover generalized compositional patterns, integrate redundancy-aware rewards to suppress category repetition, and investigate multimodal reasoning models as richer bundle-level reward evaluators.

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